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INFORMATION TECHNOLOGY DEPARTMENT

IT 343A: IT Electives

Educational Tour Journal

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DAY 1

VST ECS PHILS., INC.

1. What specific future trend did you see in active operation at this site?

During our visit to VST ECS Philippines, Inc., I observed several emerging technologies that are already being implemented in educational and business environments. These included high-performance ASUS laptops, interactive display panels, and Huawei smart boards equipped with built-in cameras and collaboration tools. These technologies support digital transformation, smart classrooms, and hybrid learning environments, which are expected to become even more prevalent in the coming years.

2. How does this agency/industry use IT to solve a specific real-world problem or improve their service delivery?

VST ECS serves as a technology solutions provider that helps educational institutions, businesses, and organizations improve productivity and communication. Their smart devices and interactive technologies enhance teaching, presentations, virtual meetings, and collaborative work. By integrating modern hardware and software solutions, they help organizations adapt to the increasing demand for digital learning and remote collaboration.

3. Did you observe any potential risk or ethical challenges in their current system?

One challenge I observed is the high cost associated with implementing advanced technological solutions, which may limit accessibility for smaller institutions or organizations with limited budgets. Additionally, since many devices are connected to networks and cloud services, cybersecurity risks and data privacy concerns must be carefully managed to protect sensitive information.

4. Based on what you saw today, how do you predict this specific agency's IT infrastructure will change by the year 2030?

By 2030, I believe VST ECS will continue to expand its portfolio of AI-powered and cloud-integrated technologies. Smart classrooms may become fully interactive, utilizing artificial intelligence for personalized learning experiences. Furthermore, collaborative technologies will likely become more immersive through augmented reality (AR), virtual reality (VR), and intelligent automation.

UNTV

1. What specific future trend did you see in active operation at this site?

At UNTV, I observed the ongoing digital transformation of the broadcasting industry. Modern broadcasting technologies, digital media platforms, and automated production systems are being utilized to deliver information efficiently across multiple channels.

2. How does this agency/industry use IT to solve a specific real-world problem or improve their service delivery?

UNTV uses information technology to collect, process, manage, and distribute information quickly and accurately. Through digital broadcasting systems, content management platforms, and online streaming services, they can provide news, public service announcements, and educational content to a wider audience in real time.

3. Did you observe any potential risk or ethical challenges in their current system?

One major challenge is ensuring the accuracy and credibility of information before it is broadcast to the public. In addition, cybersecurity threats such as unauthorized access, data breaches, and misinformation can potentially affect the reliability of media services.

4. Based on what you saw today, how do you predict this specific agency's IT infrastructure will change by the year 2030?

I anticipate that UNTV will adopt more AI-driven technologies for content creation, audience engagement, and media analytics. Future broadcasting platforms may become more personalized, interactive, and data-driven, allowing audiences to receive customized content based on their preferences.

Daily Reflection Log – The Daily Synthesis

The first day of our educational tour was both informative and inspiring. At VST ECS, I was amazed by the advanced technologies designed to improve learning and collaboration. Seeing these devices firsthand helped me understand how technological innovation can transform educational environments and workplace communication.

Our visit to UNTV provided valuable insights into the role of information technology in modern media and broadcasting. I learned how digital systems enable organizations to deliver timely and reliable information to the public. These experiences broadened my understanding of the diverse applications of IT and introduced me to potential career paths beyond software development and networking.

DAY 2

HYTEC POWER INC.

1. What specific future trend did you see in active operation at this site?

At HYTEC Power Inc., I observed robotics, drone technology, automation systems, and Arduino-powered robotic platforms. These technologies represent the growing trend of automation and intelligent systems that are increasingly being adopted across industries.

2. How does this agency/industry use IT to solve a specific real-world problem or improve their service delivery?

HYTEC utilizes advanced technologies and training equipment to educate students, educators, and professionals in robotics and automation. Through hands-on learning experiences, trainees can develop practical skills that are highly relevant in today's technology-driven workforce.

3. Did you observe any potential risk or ethical challenges in their current system?

As automation continues to evolve, some traditional jobs may become partially automated, creating concerns about workforce displacement. This highlights the importance of continuous learning, upskilling, and adapting to technological advancements.

4. Based on what you saw today, how do you predict this specific agency's IT infrastructure will change by the year 2030?

By 2030, HYTEC may incorporate more artificial intelligence, machine learning, and virtual simulation technologies into its training programs. Advanced robotics laboratories and immersive learning environments may become standard components of technical education.

PNP COMMAND CENTER

1. What specific future trend did you see in active operation at this site?

I observed the use of smart surveillance systems, integrated communication technologies, and real-time data analytics to support law enforcement operations and public safety.

2. How does this agency/industry use IT to solve a specific real-world problem or improve their service delivery?

The PNP Command Center utilizes information systems to monitor incidents, coordinate emergency responses, manage communication among units, and facilitate rapid decision-making. These technologies enhance operational efficiency and improve public safety services.

3. Did you observe any potential risk or ethical challenges in their current system?

> The extensive use of surveillance technologies raises concerns regarding privacy, data protection, and responsible information management. Appropriate policies and safeguards are necessary to ensure ethical use of collected data.

4. Based on what you saw today, how do you predict this specific agency's IT infrastructure will change by the year 2030?

> I expect future law enforcement systems to incorporate AI-assisted monitoring, predictive analytics, facial recognition enhancements, and advanced emergency response technologies that can help improve crime prevention and public safety initiatives.

Daily Reflection Log – The Daily Synthesis

The second day was one of the most exciting parts of the educational tour. At HYTEC Power, I was fascinated by the drones, robots, and automation technologies on display. Seeing these innovations in action inspired me to further explore robotics and emerging technologies.

The visit to the PNP Command Center demonstrated how information technology can contribute to public safety and emergency response. I gained a deeper appreciation for how real-time information systems support decision-making and improve operational effectiveness. It also reminded me of the importance of balancing technological advancement with ethical considerations such as privacy and security.

DAY 3

MARIANO MARCOS STATE UNIVERSITY (MMSU)

1. What specific future trend did you see in active operation at this site?

> One future trend I observed was the use of intelligent, data-driven decision support systems in education. The university demonstrated a system that analyzes student performance and recommends academic programs that align with their strengths and capabilities.

2. How does this agency/industry use IT to solve a specific real-world problem or improve their service delivery?

> MMSU uses technology to support student success by providing guidance in academic decision-making. Their systems help students identify programs that match their abilities, potentially improving academic performance and career readiness. The university's modern IT facilities also reflect its commitment to technological advancement.

3. Did you observe any potential risk or ethical challenges in their current system?

> A potential concern is the possibility of students relying too heavily on automated recommendations. While such systems can provide valuable guidance, students should still maintain the freedom to pursue their personal interests and career aspirations.

4. Based on what you saw today, how do you predict this specific agency's IT infrastructure will change by the year 2030?

> I believe MMSU will continue integrating artificial intelligence, learning analytics, and smart educational technologies. Future campuses may feature fully connected smart classrooms, advanced research facilities, and personalized digital learning platforms.

Daily Reflection Log – The Daily Synthesis

As a student, I found the MMSU visit particularly meaningful. The program recommendation system demonstrated how data and technology can positively influence educational outcomes. I was also impressed by their modern IT building, which symbolizes the growing importance of technology in higher education. The experience motivated me to continue developing my skills and preparing for the future of the IT industry.

DAY 4

DICT ILOCOS SUR

1. What specific future trend did you see in active operation at this site?

At DICT, I observed the growing implementation of digital governance and nationwide connectivity initiatives. The eGov App and community internet programs showcase how government services are becoming increasingly digital and accessible.

2. How does this agency/industry use IT to solve a specific real-world problem or improve their service delivery?

DICT leverages information technology to simplify public access to government services through digital platforms. They also address connectivity challenges by providing free public Wi-Fi and deploying specialized mobile units that bring internet access to underserved communities.

3. Did you observe any potential risk or ethical challenges in their current system?

Since government systems handle large amounts of personal information, ensuring data privacy, cybersecurity, and secure digital transactions is critical. Another challenge is bridging the digital divide to ensure equitable access to technology across all communities.

4. Based on what you saw today, how do you predict this specific agency's IT infrastructure will change by the year 2030?

I foresee DICT expanding digital government services and strengthening national connectivity infrastructure. More public services may become fully digital, enabling faster transactions, increased efficiency, and improved citizen engagement.

Daily Reflection Log – The Daily Synthesis

The DICT visit helped me appreciate how information technology can improve the lives of entire communities. Their efforts to provide internet connectivity and digital services demonstrate the significant role technology plays in national development. This experience showed me that IT is not only about creating systems but also about empowering people through access to information and services.

DAY 5

BAGUIO SMART CITY COMMAND CENTER

1. What specific future trend did you see in active operation at this site?

The most prominent trend I observed was the implementation of Smart City technologies, including integrated surveillance systems, traffic monitoring solutions, and real-time data analytics. These technologies enable more efficient city management and decision-making.

2. How does this agency/industry use IT to solve a specific real-world problem or improve their service delivery?

The command center uses information technology to monitor traffic conditions, support public safety initiatives, coordinate emergency responses, and manage city operations. Centralized access to real-time information allows city officials to respond more effectively to various situations.

3. Did you observe any potential risk or ethical challenges in their current system?

One of the primary concerns is data privacy, as smart city technologies rely heavily on collecting and processing large volumes of information. Ensuring transparency, accountability, and proper data protection measures is essential.

4. Based on what you saw today, how do you predict this specific agency's IT infrastructure will change by the year 2030?

I believe the command center will become increasingly AI-driven, utilizing predictive analytics, automated incident detection, intelligent traffic management systems, and advanced data integration platforms to improve urban services and public safety.

Daily Reflection Log – The Daily Synthesis

The Baguio Smart City Command Center was one of the most memorable destinations during our educational tour. It demonstrated how information technology can directly improve urban living by making cities safer, smarter, and more efficient. Observing these systems in operation strengthened my interest in smart technologies, data analytics, and innovative IT solutions. The experience inspired me to consider how future IT professionals can contribute to building smarter and more connected communities.